

🌀 M002 Week 1 — While Loop

Topic: Repeat Until a Wall · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

A while loop repeats. It checks first, then moves. It stops at a wall.

1 · Trace It 🕒

```
while no wall ahead:  
    move forward
```

The wall is 4 steps away. How many times does the agent move?

2 · Real Code

This is the real check.

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

`agent.detect(BLOCK, FORWARD)` is true when a wall is ahead.

Fill the blanks:

```
while not agent.detect(BLOCK, ____):  
    agent.____(FORWARD, 1)
```

3 · Fix the Bug 🐛

The agent should move until the wall. This is backwards.

```
while agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

Write the fix:

4 · Write It

Make the agent move forward until a wall, then turn right.

Use `while` , `agent.detect` , `agent.move` , `agent.turn` .

5 · Finish the Maze 📷

Open your homework world. Use a while loop. Move the agent down the straight maze to the wall.

Send a photo OR video of the agent at the wall. Then write 2-3 sentences.

My loop checked ...

The agent stopped because ...

A while loop helps because ...

6 · Video Plan

Record on a phone while the agent runs. Use: **while**, **loop**, **detect**, **wall**.

1. Run your code.
2. Read your loop. Say what stops it.
3. Say why a loop beats many **move** lines.

Plan your words:

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.